

Adriana Felisa Chávez De la Peña

Postdoctoral Scholar at Stanford Graduate School of Education

Curriculum Vitae

Contact Information

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I am a cognitive scientist and quantitative psychologist interested in **cognitive psychometrics**. I have a strong background in **cognitive psychology**, **statistics**, **Bayesian methods**, **computational approaches**, and **psychometrics**. During my Ph.D. training, I focused on facilitating the Bayesian implementation of sequential sampling models (i.e., cognitive process models for choice and response time data) to be used as measurement models through the development of software (e.g., custom models in JAGS), statistical methods (e.g., “EZ” sampling distributions) and tutorials (i.e., sample applications). I currently have a postdoctoral appointment at the Graduate School of Education at Stanford University, where I work with Dr. Jelena Obradović on a project focused on developing robust, scalable measurement tools to study executive functions, creativity, and exploration in young children.

RESEARCH INTERESTS

- Cognitive Psychometrics
- Bayesian cognitive modeling
- Individual differences
- Latent variable models
- Sequential sampling models
- Response time modeling
- Structural equation modeling
- Measurement models

SKILLS

Statistical Methods: Formal training and extensive experience in frequentist and Bayesian statistical methods for data analysis, inference, and model evaluation.

Modeling: Broad experience in the specification and implementation of Bayesian cognitive models for data analysis and hypothesis testing.

Computational Skills: Advanced R programming. Python and MATLAB programming. Linux. Version control with git. Virtual machines for code reproducibility.

Psychometrics: Broad experience in test development and validation. Field experience in the implementation of psychometric models to inform decision-making.

Teaching: Extensive experience as a teaching assistant and main instructor on courses and workshops covering technical materials at various levels.

LANGUAGES

Spanish (Native), **English** (Fluent).

EDUCATION

Ph.D. in Cognitive Sciences 2026
University of California, Irvine: Department of Cognitive Sciences

M.S. in Statistics 2025
University of California, Irvine: Department of Statistics

B.A. in Psychology 2018
National Autonomous University of México: School of Psychology

RESEARCH EXPERIENCE

All code and materials for my research projects are openly available at github.com/Adrifelcha.

Postdoctoral Scholar

Graduate School of Education
Stanford University

P.I.: Jelena Obradović

2026 - Present
Stanford, California

Graduate Research Assistant

Cognition and Individual Differences Lab
University of California, Irvine
Department of Cognitive Sciences

P.I.: Joachim Vandekerckhove

2022 - 2025
Irvine, California

Graduate Research Assistant

Structure in Perception and Cognition Lab
University of California, Irvine
Department of Cognitive Sciences

P.I.: Jeffrey N. Rouder

2020 - 2022
Irvine, California

Undergraduate Research Assistant

Learning and Adaptive Behavior Lab
National Autonomous University of México
School of Psychology

P.I.: Arturo Bouzas Riaño

2015 - 2018
México City, México

ACADEMIC SERVICE

Peer reviewer

Research in Developmental Disabilities

Virtual Conference Moderator

Society for Mathematical Psychology
Annual Virtual Meetings

2021 - 2025

Colloquium Committee

University of California, Irvine
Department of Cognitive Sciences

2023 - 2024

Elected student representative

National Autonomous University of México
School of Psychology
Psychology School Technical Council

2014 - 2016

TEACHING EXPERIENCE

Instructor positions

University of California, Irvine Research Methods (COGS 112M)	2025 - 2026 Irvine, California
University of California, Irvine Probability and Inference (PSYCH/COGS 10B)	Summer, 2023 Online course
Jean Piaget Educational Center (Private Highschool) Research methods I, II and III	2018 - 2020 México city, México

Teaching Assistant positions

University of California, Irvine School of Social Sciences Probability and inference: 10A, 10B and 10C Psychology fundamentals: 9A, 9B and 9C	2020 - 2025 Irvine, California
National Autonomous University of México School of Psychology Learning and Adaptive Behavior: ACA-I, II and III	2016 - 2018 México City, México

Summer Workshops at the National Autonomous University of México

• <i>Teoría de la Probabilidad para Psicólogos</i> <i>Probability Theory for Psychologists</i>	June 2018
• <i>Fundamentos de la Estadística Inferencial para las Ciencias Sociales</i> <i>Inferential Statistics fundamentals for Social Sciences</i>	June 2018
• <i>Introducción al Pensamiento Estadístico</i> <i>Introduction to Statistical Thinking</i>	June 2017
• <i>Introducción al Modelamiento Bayesiano en Ciencias Cognitivas</i> <i>Introduction to Bayesian Modeling in Cognitive Sciences</i>	June 2017
• <i>Un Laboratorio Virtual en Python para Modelos del Comportamiento</i> <i>A virtual lab for cognitive modeling in Python</i>	June 2017

MENTORING EXPERIENCE

Senior graduate student mentor Cognition and Individual Differences lab Mentoring undergraduate students on their honors theses	2024 - 2025 Irvine, California
Undergraduate software developer Learning and Adaptive Behavior Lab Led project "Development of Virtual tools for teaching mathematical models in Psychology"	2016 - 2018 México City, México
Senior undergraduate mentoring program Learning and Adaptive Behavior Lab Undergraduate mentor on coding and experimental design for the students enrolled in the "Research workshop" taught by Arturo Bouzas Riaño	2015 - 2018 México City, México

PUBLICATIONS

Published papers

- **Chávez De la Peña, A.F.**, Shin, E., & Vandekerckhove, J. (2026). Robust Bayesian hypothesis testing with the hierarchical EZ-DDM. *Behavior Research Methods*, 58, 177.
- **Chávez De la Peña, A.F.**, & Vandekerckhove, J. (2025). An EZ Bayesian hierarchical drift diffusion model for response time and accuracy. *Psychonomic Bulletin & Review*, 32, 3067–3087.
- Etz, A., **Chávez De la Peña, A.F.**, Baroja, L., Medriano, K., & Vandekerckhove, J. (2024). The HDI + ROPE decision rule is logically incoherent but we can fix it. *Psychological Methods*.
- Villarreal, J., **Chávez De la Peña, A.F.**, Mistry, P., Menon, V. E., Vandekerckhove, J., & Lee, M. D. (2024). Bayesian graphical modeling with the circular drift diffusion model. *Computational Brain & Behavior*, 7, 181-194.

Pre-print

- Rouder, J., **Chávez De la Peña, A.F.**, Mehrvarz, M., & Vandekerckhove, J. (preprint). On Cronbach's merger: Why experiments may not be suitable for measuring individual differences. *PsyArxiv*.

Theses

- **Chávez De la Peña, A.F.** (2026). *Modeling individual differences in choice and response time*. Ph.D. thesis.
- **Chávez De la Peña, A.F.** (2018). *Estudios con deteccion de señales (Studies on Signal Detection)*. Undergraduate thesis.

PRESENTATIONS

Conference presentations

- **Chávez De la Peña, A.F.** and Joachim Vandekerckhove “Robust, scalable Cognitive Psychometrics with the Hierarchical Bayesian EZ-DDM.” 2026 Meeting of the Society for Mathematical Psychology. Montreal, Canada. July, 2026. Presented by J. Vandekerckhove.
- **Chávez De la Peña, A.F.** and Joachim Vandekerckhove “An EZ Bayesian hierarchical drift diffusion model for response time and accuracy.” 2024 Meeting of the Society for Mathematical Psychology. Tilburg, Netherlands. July, 2024
- **Chávez De la Peña, A.F.** “Modeling Individual Differences with an EZ Bayesian hierarchical drift diffusion model.” Bayesian Cognitive Modeling Talk series hosted by the Learning and Adaptive Behavior Lab at the School of Psychology, UNAM. México City, México. March, 2024
- **Chávez De la Peña, A.F.**, Villarreal Ulloa Manuel, Michael D. Lee and Joachim Vandekerckhove “A Bayesian hierarchical implementation of the circular drift diffusion model.” 2023 Meeting of the Society for Mathematical Psychology. Amsterdam, Netherlands. July, 2023
- **Chávez De la Peña, A.F.** “Drifting beyond the Bayesics.” 58th Annual Edwards Bayesian Research Conference. Fullerton, California. March, 2023.
- **Chávez De la Peña, A.F.**, Jeffrey N. Rouder and Joachim Vandekerckhove “Principal-component exploration of individual differences in the general-speed component of response times.” 2022 Meeting of the Society for Mathematical Psychology. Toronto, Canada. July, 2022.

- **Chávez De la Peña, A.F.**, Jeffrey N. Rouder and Joachim Vandekerckhove “*Principal-component exploration of individual differences in the general-speed component of response times.*” 2022 Virtual Meeting of the Society for Mathematical Psychology. July, 2022.
- **Chávez De la Peña, A.F.** “*El Efecto Espejo en Percepción: No es otro estudio de Memoria de Reconocimiento.*” VI International Seminar of Behavior and Applications. Tlaxcala, México. November, 2017

Keynote and invited presentations

- **Chávez De la Peña, A.F.** “*Cognitive Psychometrics: Building Bayesian cognitive measurement models.*” Invited talk at the Berkeley Evaluation & Assessment Research Center (BEAR seminar series). University of California, Berkeley, California. February, 2026
- **Chávez De la Peña, A.F.** “*Modelamiento Bayesiano con el Modelo de Difusion Circular.*” Keynote presentation at the School of Psychology at the National Autonomous University of México. México City, México. September, 2024

Posters

- **Chávez De la Peña, A.F.**, Eunice Shin and Joachim Vandekerckhove “*Robust Bayesian hypothesis testing with the hierarchical EZ-DDM.*” 2025 Meeting of the Psychonomic Society. Denver, Colorado, United States. November, 2025.
- **Chávez De la Peña, A.F.**, Eunice Shin and Joachim Vandekerckhove “*EZ Cognitive Psychometrics: Bayesian hypothesis testing with the EZ-DDM.*” 2025 Meeting of Society for Mathematical Psychology. Columbus, Ohio, United States. July, 2025.
- **Chávez De la Peña, A.F.** and Joachim Vandekerckhove “*An EZ Bayesian hierarchical drift diffusion model for response time and accuracy.*” 2024 Meeting of the Psychonomic Society. New York, United States. November, 2024.
- **Chávez De la Peña, A.F.**, Rouder, J. and Joachim Vandekerckhove “*Individual differences in the general speed component of Response Times.*” Symposium of Individual Differences in cognition. San Francisco, California. November, 2023.
- **Chávez De la Peña, A.F.**, Rouder, J., and Joachim Vandekerckhove “*Exploring the uni-factorial structure of the general-speed component of Response Times.*” 7th Summer School of Computational and Mathematical Modeling of Cognition. Szklarska Poreba, Poland. July, 2022.
- Ruiz, N.K., Niño, J.M., and **Chávez De la Peña, A.F.** “*Studies on Framing Effect: Language compatibility.*” 40th Annual Conference of the Society for Judgment and Decision Making. Montreal, Canada. November, 2019.
- Niño, J.M., Villalobos, E., Villarreal, J.M., **Chávez De la Peña, A.F.** and Lira, F. “*Transitivity in gains and losses.*” 40th Annual Conference of the Society for Judgment and Decision Making. Montreal, Canada. November, 2019.
- **Chávez De la Peña, A.F.**, Lee M. D. and Bouzas, A. “*Bayesian Cognitive and Statistical Modeling Applied to Signal Detection Theory and the Mirror Effect in a Perceptual Task.*” 52nd Annual Meeting of the Society for Mathematical Psychology. Montreal, Canada. July, 2019.
- **Chávez De la Peña, A.F.** “*The Mirror Effect within Perception: Not another Recognition Memory study.*” Object Perception Attention and Memory Meeting. Boston, Massachusetts. November, 2016.
- **Chávez De la Peña, A.F.** “*La Sensibilidad como fuente de Sesgo en una tarea de detección de señales usando la ilusión de Ebbinghaus*” (“*Sensibility as a source of Bias in a signal detection task using the Ebbinghaus illusion*”). V International Seminar on Behavior and Applications. México City, México. November, 2015.

HONORS & AWARDS

- People's Choice Award: Best Poster** [2025]
 2025 Meeting for the Society for Mathematical Psychology
 Title: EZ Cognitive Psychometrics:
 Bayesian hypothesis testing with the EZ-DDM
- Graduate Travel Award** [2024]
 Psychonomics Society
- Travel Award** [2024]
 Society for Mathematical Psychology
- People's Choice Award: Best talk** [2022]
 Virtual meeting for the Society for Mathematical Psychology
 Title: Principal-component exploration of individual differences
 in the general-speed component of response times.

MEMBERSHIPS

- Member of the Psychonomic Society 2023 - Present
 Member of the Society for Mathematical Psychology 2019 - Present

SCHOLARSHIPS

- UCMEXUS-CONACYT** 2021 - 2025
 Scholarship for Mexican Ph.D. students abroad
 Granted by the National Commission of Science and Technology (México)
 supported by the University of California (UCMEXUS)
- Exchange Abroad Program** Fall, 2014
 From: National Autonomous University of México
 To: University of California Santa Barbara
 Sponsored by Carlos Slim's foundation.

SUMMER SCHOOLS

- 6th European Summer School on Computational and Mathematical Modeling of Cognition** July, 2022
 Szklarska Poreba, Poland
 Instructors: Stephan Lewandowsky, Jana Jarecki, Michael Nunez, Klaus Oberauer, Gordon Brown, Cas Ludwig, Chris Donkin, Joachim Vandekerckhove, Laura Fontanesi, Beatrice Kuhlman.

OTHER PROFESSIONAL EXPERIENCE

Test Development Coordinator

January - September, 2020

National Autonomous University of México

México City, México

Supervisor: Ramses Vázquez Lira

I coordinated the work of 13 teams in charge of designing items and tasks to be used in different measurement instruments considered as part of the Assessment System used to inform hiring and promotion decisions within the public education system. Teams were comprised of specialists in test development, experienced teachers and/or principals, and substantive experts.

Psychometrician - Data Analyst

2019 - 2020

Applied Cognitive Diagnostics (start-up)

México City, México

Supervisor: Ramses Vázquez Lira

I worked on various small projects where private companies would request us to collect and analyze data regarding the application of different measurement instruments (e.g., to keep track of their employees' stress levels and mental well-being in compliance with current government regulations).

Psychometrician - External consultant

September - December, 2019

Secretariat of Public Education

México City, México

Supervisor: Ramses Vázquez Lira

I was part of a small group of consultants who helped design the new Assessment System to inform hiring and promotion decisions in the public education system. We defined the sets and types of measurement instruments that should be considered for different stages and positions in order to capture traits of interest.

Psychometrician - Data Analyst

January - September, 2019

International Network for Educational Assessment

México City, México

Supervisor: Ramses Vázquez Lira

I analyzed data collected across different schools/states/districts in different applications of the National Protocol for Learning Assessment (i.e., an exam known as PLANEA), to deliver detailed reports on the academic performance to the interested parties (e.g., private schools, state government, etc.).

Psychometrician - Test Validation

May - December, 2018

National Institute for Educational Evaluation

México City, México

Department of Assessment of Teachers and Principals

Supervisor: Sandra Conzuelo Serrato

I was hired as the head psychometrician of the Basic Education Division of the Department of Assessment of Teachers and Principals in the public education system. My job was to verify the validity and reliability of the measurement instruments used to assess the performance of teachers, principals and supervisors in the national public education system.

Internship at Private Educational Center

June - November, 2013

Association of Behavioral Science Specialists

México City, México

Supervisor: Alma Hernandez Mendoza

I did a short internship at a private center for neurodivergent children and children with learning difficulties where I applied and interpreted standardized psychometric tests to identify the specific needs of every student, so that the education specialists could design a learning plan to be followed up by their personal tutors.